VITyarthi Program

MASTER PROJECT REPORT

Digital Literacy, Communication & Cybercrime Awareness

Task	Title	Deliverables
1	Introduction to Digital Literacy	Reflection on digital skills, tools & responsible use
2	Evaluating Online Information	Source credibility, misinformation analysis, SIFT method
3	Online Privacy & Data Security	Privacy settings, data footprint, password hygiene
4	Email Etiquettes & Social Media Responsibility	Professional emails, Do's & Don'ts checklist
5	Cybercrime Awareness	Phishing case study, prevention checklist, personal reflection

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TASK 1

Introduction to Digital Literacy

Understanding digital tools, skills, and responsible online behaviour

1.1 Overview

Digital literacy refers to the ability to use digital technologies effectively, critically, and responsibly. As a third-year CSE student specialising in AI & ML, digital tools are central to both academic learning and future professional work. This task established the foundational vocabulary and mindset for the rest of the VITyarthi program.

1.2 Key Concepts Covered

- Definition of digital literacy and why it matters for 21st-century professionals.
- The distinction between being digitally connected and being digitally competent.

- Core digital skills: information literacy, communication, content creation, safety, and problem-solving.
- Responsible digital citizenship — understanding the ethical and social dimensions of online behaviour.
- The concept of a digital footprint and how every online action contributes to a permanent, searchable record.

1.3 Reflection

The most significant takeaway from Task 1 was recognising that technical ability and digital literacy are not the same thing. A student can be proficient in coding or data science and still lack the critical thinking skills needed to evaluate online information, communicate professionally, or protect their personal data. This distinction shaped the approach to all subsequent tasks.

TASK 2 — Evaluating Online Information

Source credibility, misinformation detection, and the SIFT method

2.1 Overview

The internet is simultaneously the world's largest library and its largest source of misinformation. Task 2 focused on developing the critical thinking skills necessary to distinguish reliable information from false or misleading content — a skill that is directly relevant to AI and ML research, where training data quality is paramount.

2.2 The SIFT Method

The SIFT framework provides a structured approach to evaluating any piece of online information:

Letter	Stands For	What To Do
S	Stop	Pause before sharing or believing. Ask: do I know this source? Am I in an emotional state that might cloud my judgment?
I	Investigate the Source	Look up the source itself before reading the content. Who published this? What is their purpose? Do they have known biases?
F	Find Better Coverage	Search for the same claim across multiple reputable sources. If only one outlet is reporting it, treat it with scepticism.

T	Trace Claims, Quotes & Media	Go upstream. Find the original source of a claim, quote, or image rather than relying on a secondary report of it.

2.3 Types of Misinformation

- Misinformation — false content shared without intent to deceive (e.g., forwarded WhatsApp messages).
- Disinformation — deliberately false content created to mislead (e.g., fabricated news articles).
- Malinformation — true information used with harmful intent (e.g., leaked private photos).
- Deepfakes — AI-generated audio/video content that falsely depicts real people saying or doing things they did not.

2.4 Credibility Indicators

Reliable sources typically share these characteristics: clear authorship with verifiable credentials, transparent editorial or funding disclosures, citations to primary data or peer-reviewed research, and a track record of corrections when errors occur. Applying these criteria is especially important when sourcing data for AI models, where garbage-in means garbage-out.

TASK 3 — Online Privacy & Data Security

Managing your digital footprint, passwords, and personal data

3.1 Overview

Every digital action — a search query, a location check-in, an app permission grant — generates data. Task 3 examined how that data is collected, stored, and potentially exploited, and what practical steps individuals can take to protect their privacy in an era of pervasive digital surveillance.

3.2 Understanding Your Digital Footprint

A digital footprint has two components:

- Active footprint — data you deliberately share: social media posts, form submissions, emails, profile information.
- Passive footprint — data collected without explicit action: browsing history, location data, device identifiers, cookies, and app usage patterns.

For college students, passive footprints are often far larger than they realise. Every app installed, every permission granted, and every free service used is an exchange of personal data for functionality.

3.3 Password Hygiene — Best Practices

Good Practice	Why It Matters
Use passwords of 12+ characters with mixed types	Longer, complex passwords are exponentially harder to brute-force.
Never reuse passwords across accounts	One compromised password should not unlock all your accounts.
Use a password manager (Bitwarden, Google)	Allows unique, complex passwords for every account without memorising them.
Enable Two-Factor Authentication (2FA) everywhere	A stolen password is useless if the attacker also needs your phone's OTP.
Change passwords immediately if a service reports a breach	Data breaches are common; acting fast limits exposure.

3.4 Privacy Settings Audit

A regular privacy audit should cover: social media visibility settings (who can see posts, tags, and contact details), app permissions (camera, microphone, location, contacts), browser cookies and tracking settings, and account recovery options (keep backup emails and phone numbers current).

TASK 4 — Email Etiquettes & Social Media Responsibility

Professional emails, digital communication, and responsible online conduct

4.1 Overview

Task 4 had two parts: writing professional emails following correct etiquette (Part A), and developing a social media responsibility checklist for college students (Part B). Additionally, a critical reflection was written on a scenario where poor digital communication caused real harm.

4.2 Part A — Professional Email Writing

Email 1: Extension Request to Professor

Field	Detail

From	aryan123@gmail.com
To	narendrasing@gmail.com (Professor)
Date	March 23, 2026
Subject	Extension Request — Static Equilibrium Assignment

The email requested a 3-day extension on a Static Equilibrium and Vector Forces assignment due to illness. Key etiquette elements demonstrated:

- Formal salutation and warm opening sentence.
- Specific assignment name and current due date mentioned upfront.
- Brief, honest reason without oversharing personal medical details.
- Concrete alternative deadline proposed (26th March, 5:00 PM).
- Attachment of in-progress work to demonstrate good faith.
- Polite closing with student ID for identification.

Email 2: Summer Internship Interest — Reliance

Field	Detail
From	aryan123@gmail.com
To	maheshpatel@gmail.com (Internship Coordinator, Reliance)
Date	March 23, 2026
Subject	Summer Internship Opportunities

The internship email expressed formal interest in a summer engineering position at Reliance, referencing the company's sustainable energy initiatives. Etiquette elements:

- Professional self-introduction with institution and year of study.
- Company-specific reference (sustainable energy) showing genuine research.
- Relevant skill highlighted — CAD modelling for energy infrastructure.
- Resume and portfolio attached — complete, professional application package.
- Open-ended closing inviting dialogue without pressure.

4.3 Part B — Social Media Responsibility: Do's and Don'ts

DO's	DON'Ts
Audit privacy settings regularly to control who sees your posts and tagged photos.	Post confidential information such as student ID, home address, or research project details.

Use a professional profile picture and clear bio on LinkedIn to build a positive digital brand.	Engage in cyber-venting or posting heat-of-the-moment complaints about professors or peers.
Verify information before sharing to prevent the spread of misinformation.	Plagiarize or share copyrighted material without proper attribution.
Network with industry professionals and alumni through relevant groups and discussions.	Post content depicting illegal activities or policy violations.
Think before tagging others in photos; respect everyone's privacy and professional boundaries.	Ignore your digital footprint — assume deleted content is permanently gone.

4.4 Critical Reflection — Poor Digital Communication Scenario

The Scenario (Hypothetical)

A final-year engineering student, furious after receiving an internship rejection, posted a public rant on Instagram tagging the company's HR representative and a classmate by name, calling the selection process 'rigged and corrupt.' The post remained public for 18 hours. Consequences: the professor who had been considering the student for a research role withdrew the offer; a formal complaint was filed by the tagged peer; the student received a written academic warning; and three other internship applications were rejected after recruiters found screenshots of the post circulated by mutual contacts. Deleting the post changed nothing — the screenshots had already spread.

What could have been done differently:

- Draft, pause, and reflect — wait 24 hours before posting anything written in anger.
- Use a private channel — express frustration to a trusted friend, not a public platform.
- Send a professional follow-up email to the company requesting feedback.
- Remember permanence — screenshots preserve deleted content indefinitely.
- Never tag real individuals in a complaint — it escalates a personal grievance into a legal matter.

TASK 5

Cybercrime Awareness

Phishing case study, Stay Safe Online checklist, and personal reflection

5.1 Part A — Case Study: Phishing

Phishing is a form of digital fraud in which attackers impersonate trusted entities — banks, government agencies, or service providers — to steal credentials, financial data, or personal identification. Unlike traditional hacking, phishing targets human psychology rather than software vulnerabilities.

Attack Stage	Description

Preparation	The attacker clones a legitimate organisation's branding — logos, email addresses, fonts — to create a convincing fake email or website.
The Hook	A mass message is sent with an urgent call to action: 'Your KYC is incomplete,' 'Account compromised,' or 'Claim your scholarship.' Urgency prevents critical thinking.
Redirection	The victim clicks a malicious link leading to a spoofed login page identical to the real site (e.g., a fake SBI or Google login).
Theft	As the victim enters credentials or OTP, the attacker captures data in real time on a server they control.
Exploitation	Stolen data is used to drain accounts, sell on the dark web, hijack social profiles, or launch further attacks on the victim's contacts.

Who Is Targeted and Consequences

- Financial loss — bank accounts, UPI wallets, and credit cards drained within minutes.
- Identity theft — personal details used to open fraudulent loans or credit cards.
- Academic harm — hijacked college portal or email accounts used to manipulate records.
- Reputational damage — compromised social media used to spread offensive content.
- Psychological impact — lasting anxiety, shame, and distrust of digital systems.

5.2 Part B — Stay Safe Online: Prevention Checklist

The following checklist was created specifically for college students in India and saved as prevention-checklist.md inside the task-5-cybercrime/ folder of the GitHub repository.

#	Tip	Action Required
1	**Strong Unique Passwords**	Use uppercase, lowercase, numbers & symbols. Never reuse passwords. Use a password manager like Bitwarden.
2	**Enable 2FA on All Accounts**	Two-Factor Authentication ensures a stolen password alone cannot unlock your accounts.
3	**Never Click Unknown Links**	Verify URLs before clicking messages about scholarships, job offers, or urgent account warnings.
4	**Keep Devices & Apps Updated**	Updates patch security vulnerabilities — enable auto-updates for OS, browser, and apps.
5	**Limit Personal Info on Social Media**	Do not post phone number, address, college ID, or travel plans publicly.
6	**Use Secure Wi-Fi or VPN**	Avoid banking apps on public Wi-Fi. Use a VPN if you must connect to open networks.
7	**Verify Authority Claims**	No legitimate official demands instant payment or OTP over WhatsApp or phone.
8	**Audit App Permissions**	Revoke unnecessary access. A flashlight app does not need your contacts.

9 ★	**Never Share UPI PIN or OTP**	Banks will NEVER ask for your PIN or OTP. OTP is for authorising payments — sharing it is losing money. (UPI Safety)
10 ★	**QR Codes Cannot Receive Money**	Scanning a QR code SENDS money. 'Scan to receive' is always a scam. Share your UPI ID to receive. (UPI Safety)

National Cyber Crime Portal	National Cyber Helpline
cybercrime.gov.in	**1930 (24x7, Free)**

5.3 Personal Reflection (100–150 Words)

What surprised me most, and the habit I will change — Govind Jadhav, 25BAI11219

What surprised me most while researching phishing was how little technical skill attackers actually need. I had assumed cybercrime required deep programming knowledge, but phishing kits — ready-made tools to clone bank and government websites — are sold on the dark web for as little as a few hundred rupees. A person with no coding ability can launch a convincing campaign targeting thousands of people within hours. That was genuinely unsettling.

The second surprise was the QR code scam. I had always assumed scanning a QR code to receive money was a normal UPI feature. Discovering that QR codes can only send money — and that the 'scan to receive' trick is one of the most common UPI frauds in India — was an eye-opener.

The one habit I will change immediately: I will stop responding to any message that creates urgency. Whether it is a scholarship alert, a bank warning, or a KYC update, I will go directly to the official website instead of clicking any link. Urgency is the attacker's greatest weapon — pausing for ten seconds is my best defence.

Word Count: 148 words

Overall Conclusion

The five tasks of the VITyarthi Digital Literacy program represent a progressive and interconnected curriculum. Task 1 established foundational awareness; Task 2 built critical thinking around online information; Task 3 addressed personal data and privacy; Task 4 developed professional communication skills; and Task 5 confronted the real and growing threat of cybercrime.

Across all five tasks, a single theme emerges: digital competence is not just about technical skill. It is about judgment — knowing when to pause before posting, when to verify before clicking, when to question before trusting, and when to report rather than ignore. For a generation that will design, build, and deploy AI systems, these habits are not optional. They are foundational.

The materials produced across these five tasks — professional emails, a social media checklist, a cybercrime prevention guide, and reflective writing — are all documented and version-

controlled in the GitHub repository: govind25bai11219-tech/govind25bai11219_VITyarthi_Project.

Repository & File Structure

Folder / File	Contents
task-4-email-etiquettes/	Email 1 (extension request), Email 2 (internship interest), Social Media Do's & Don'ts checklist
task-4-email-etiquettes/reflection.md	100–150 word reflection on poor digital communication scenario
task-5-cybercrime/	Phishing case study analysis and cybercrime awareness materials
task-5-cybercrime/prevention-checklist.md	10-point Stay Safe Online checklist with UPI safety tips and reporting channels
task-5-cybercrime/reflection.md	Personal reflection on phishing research — surprises and habit change
GitHub Repository	github.com/govind25bai11219-tech/govind25bai11219_VITyarthi_Project

Cybercrime Reporting: National Cyber Crime Portal — cybercrime.gov.in | Helpline: 1930 (24x7, Free of charge)